

1. A total of 46% of the voters in a certain city classify themselves as Independents, where as 30% classify themselves as Liberals and 24% say they are Conservatives. In s recent local election, 35% of the Independents, 62% of the Liberals and 58% of the Conservatives voted. A voter is chosen at random. Given that this person voted in the local election, the probability that he or she is an Independent is _____ [enter up to 3 decimal points]

Solution:

Given that,

$$P(I) = 46\% \quad \text{Independents}$$

$$P(L) = 30\% \quad \text{Liberals}$$

$$P(C) = 24\% \quad \text{Conservatives}$$

$\Rightarrow$ And also given that 35% of Independents are voted

$$P(V/I) = 35\%$$

$$P(V/L) = 62\%$$

$$P(V/C) = 58\%$$

$\therefore$ We have to find $P(I/V)$.

According to the Bays theorem,

$$\begin{aligned} P\left(\frac{I}{V}\right) &= P\left(\frac{P(I \cap V)}{P(V)}\right) \\ &= \frac{P\left(\frac{V}{I}\right)P(I)}{P\left(\frac{V}{I}\right)P(I) + P\left(\frac{V}{L}\right)P(L) + P\left(\frac{V}{C}\right)P(C)} \\ &= \frac{0.35 \times 0.46}{0.35 \times 0.45 + 0.62 \times 0.30 + 0.58 \times 0.24} \\ &= 0.331 \end{aligned}$$

2. Mr. Elliot has just had a biopsy or a possibly cancerous tumor. Not wanting to spoil a weekend family event, he does not want to hear any bad news in the next few days. But if he tells the doctors to call only if the news is good, then if the doctor doesn't call, Mr. Elliot can conclude that the news is bad. Being a student of probability Mr. Elliot instructs the doctor to flip a coin. If it comes up head, the doctor is to call if the new is good and not call if the new is bad. If the coin comes up tails, the doctor is not to call whether the news is good or bad.

Solution: Let α be the probability that the tumor is cancerous. Let β be the probability that the tumor is cancerous given that the doctor doesn't call then $\beta =$

(a) $\frac{a}{1+a}$

(b) $\frac{2a}{1+3a}$

(c) $\frac{1}{1+2a}$

(d) $\frac{2a}{1+a}$

Solution:

Let C be the event that tumor is cancerous. Let N be the event that the doctor doesn't call.

So we need to find $P\left(\frac{C}{N}\right)$

$$\beta = P\left(\frac{C}{N}\right) = \frac{P(C \cap N)}{P(N)} = \frac{P\left(\frac{N}{C}\right)P(C)}{P\left(\frac{N}{C}\right)P(C) + P\left(\frac{N}{C^1}\right)P(C^1)}$$

It is given that he has cancer.

If coin turns to be head $\left[p(n) = \frac{1}{2}\right]$ he wants call.

If coin turns to be tail $\left[p(t) = \frac{1}{2}\right]$ he want call.

$\therefore$ In both cases doctor won't call him $P\left(\frac{C}{N}\right) = 1$

$P\left(\frac{N}{C^1}\right)$:

It is given that he doesn't have cancer.

$\therefore$ Doctor won't call him only when the coin turns to be tail $(t) = \frac{1}{2}$

$$P\left(\frac{N}{C^1}\right) = \frac{1}{2}$$

$$P\left(\frac{C}{N}\right) = \frac{P\left(\frac{N}{C}\right)P(C)}{P\left(\frac{N}{C}\right)P(C) + P\left(\frac{N}{C^1}\right)P(C^1)}$$

$$= \frac{1 \times \alpha}{1 \times \alpha + \frac{1}{2} \times (1 - \alpha)}$$

$$= \frac{1 \times \alpha}{\alpha + \frac{1}{2}(1 - \alpha)}$$

$$= \frac{2\alpha}{2\alpha + (1 - \alpha)}$$

$$= \frac{2a}{1 + a}$$

$$\therefore \beta = \frac{2a}{1 + a}$$

3. According to the question 2, which should be bigger α or β ?

(a) $\alpha > \beta$

(b) $\alpha < \beta$

(c) $\alpha = \beta$

(d) Can't say

Solution:

$$\frac{2a}{1+a} \geq a \text{ [which is strictly inequal unless } \alpha = 1]$$

$$\beta \geq \alpha \Rightarrow \beta > \alpha$$

4. A family has children with probability P_i where $P_1 = 0.01$, $P_2 = 0.25$, $P_3 = 0.35$, $P_4 = 0.3$. A child from this family is randomly chosen. Given that this child is the eldest child in the family, the conditional probability that the family has 4 children _____.

(a) 0.24

(b) 0.18

(c) 0.26

(d) 0.16

Solution: Let E be the event the child selected is the eldest. Let F_i be the event that the family has i children. We need to find

$$P\left(\frac{F^4}{E}\right) = P \frac{(F^4 \cap E)}{P(E)}$$

$$P(F^4) = P\left(\frac{F^4}{E}\right)P(E)$$

$$= \frac{1}{4} \times 0.3$$

$$= \frac{0.3}{4}$$

$$= 0.075$$

$$P(E) = \sum P(F)P\left(\frac{E}{F_i}\right) = P(F_1)P\left(\frac{E}{F_1}\right) + P(F_2)P\left(\frac{E}{F_2}\right) + P(F_3)P\left(\frac{E}{F_3}\right) + P(F_4)P\left(\frac{E}{F_4}\right)$$

$$P(E) = 0.1 \times 1 + 0.25 \times \frac{1}{2} + 0.35 \times \frac{1}{3} + 0.3 \times \frac{1}{4}$$

$$= 0.4155$$

$$\text{Therefore, } P\left(\frac{F_4}{E}\right) = \frac{0.075}{0.4155} = 0.18$$

5. A gambler has in his pocket a fair coin and a two-head coin. He selects one of the coins at random; when he flips it, it shows head. What is the probability that it is the fair coin?

(a) $\frac{1}{2}$

(b) $\frac{1}{3}$

(c) $\frac{1}{4}$

(d) $\frac{1}{5}$

Solution:

It is given that, the coin shows head

We need to find

$$P\left(\frac{Fair}{h}\right) = \frac{P(Fair \cap h)}{P(h)}$$

$$\begin{aligned}
 &= \frac{P\left(\frac{h}{fair}\right)P(fair)}{P\left(\frac{h}{fair}\right)P(fair)+P\left(\frac{h}{not\ fair}\right)P(not\ fair)} \\
 &= \frac{\frac{1}{2} \times \frac{1}{2}}{\frac{1}{2} \times \frac{1}{2} + 1 \times \frac{1}{2}} \\
 &= \frac{\frac{1}{4}}{\frac{1}{2} + \frac{1}{2}} \\
 &= \frac{\frac{1}{4}}{\frac{3}{2}} \\
 &= \frac{1}{3}
 \end{aligned}$$

6. A gambler has a fair coin and a two headed coin. He selects one of the coins at random flipped it twice and got two heads. Now the probability that it is the fair coin is _____.

(a) $\frac{1}{2}$

(b) $\frac{1}{3}$

(c) $\frac{1}{4}$

(d) $\frac{1}{5}$

Solution:

$$\begin{aligned}
 P\left(\frac{Fair}{hh}\right) &= \frac{P(Fair \cap hh)}{P(hh)} \\
 &= \frac{P\left(\frac{hh}{fair}\right)P(fair)}{P\left(\frac{hh}{fair}\right)P(fair)+P\left(\frac{hh}{not\ fair}\right)P(no\ fair)}
 \end{aligned}$$

$$P\left(\frac{hh}{fair}\right) = 1/4[(hh), (hT), (Th), (TT)]$$

$$P\left(\frac{hh}{no\ fair}\right) = 1$$

$$P\left(\frac{Fair}{hh}\right) = \frac{\frac{1}{4} \times \frac{1}{2}}{\frac{1}{4} \times \frac{1}{2} + 1 \times \frac{1}{2}}$$

$$= \frac{\frac{1}{8}}{\frac{1}{8} + \frac{1}{2}}$$

$$= \frac{1}{5}$$

7. A gambler has a fair coin and a two headed coin. He selects one of the coins at random. He flipped the same coin thrice. In the first time it showed head, in the second time also head and in the third time it showed tails. The probability that the coin selected is fair coin is _____.

Solution:

The given sequence of outcomes is HHT. It is given that there is a tail in the outcomes which means it is not a two-head coin we can say it is fair coin

∴ Answer = 1

8. English and American spellings are "rigour" and "rigor" respectively. A man staying at a Parisian hotel writes this word, and a letter taken at random from his spelling is found to be a vowel.

If the 40% of the English speaking men at the hotel are English and 60% are Americans. What is the probability that the writer is an Englishman?

(a) $\frac{5}{13}$

(b) $\frac{4}{9}$

(c) $\frac{5}{11}$

(d) $\frac{2}{5}$

Solution:

We have to find

$$\begin{aligned}
 P\left(\frac{E}{V}\right) &= \frac{P\left(\frac{V}{E}\right)P(E)}{P\left(\frac{V}{E}\right)P(E)+P\left(\frac{V}{A}\right)P(A)} \\
 &= \frac{\frac{3}{6} \times \frac{40}{100}}{\frac{3}{6} \times \frac{40}{100} + \frac{2}{5} \times \frac{60}{100}} \\
 &= \frac{\frac{1}{2} \times \frac{2}{5}}{\frac{1}{2} \times \frac{2}{5} + \frac{2}{5} \times \frac{3}{5}} \\
 &= \frac{5}{11}
 \end{aligned}$$

9. You randomly choose a treasure chest to open and then randomly choose a coin from that treasure chest. If the coin you choose is gold then what is the probability that you choose chest A?

Treasure 1 = 100 gold coins

Treasure 2 = 50 gold coins + 50 silver coins

(a) $\frac{2}{3}$

$$(b) \frac{1}{3}$$

$$(c) \frac{1}{2}$$

$$(d) \frac{1}{4}$$

Solution:

Let G be the event of selecting a Gold coin.

$\Rightarrow$ Let A be the event selecting Treasure 1.

$\Rightarrow$ Let B be the event selecting Treasure 2.

$$P\left(\frac{G}{A}\right) = 1$$

$$P\left(\frac{G}{B}\right) = \frac{1}{2}$$

We need to find

$$\begin{aligned} P\left(\frac{A}{G}\right) &= \frac{P(A \cap G)}{P(G)} \\ &= \frac{P(A) \cdot P\left(\frac{G}{A}\right)}{P(A) \cdot P\left(\frac{G}{A}\right) + P(B) \cdot P\left(\frac{G}{B}\right)} \\ &= \frac{\frac{1}{2} \cdot 1}{\frac{1}{2} \cdot 1 + \frac{1}{2} \cdot \frac{1}{2}} \\ &= \frac{2}{3} \end{aligned}$$

10. A machine learning model M_1 predicts raining, the chance of raining is 60%, when the other model M_2 predicts raining, the chance of raining is 60%. If both predict to rain

[assuming they did the prediction independently], what is the chance of raining? [choose the most appropriate answer]

1. it will rain for sure 3. Anything between 0 and 1

2. Incomplete data 4. 36% chance

Ans:

Let X, Y, Z be the events that M1 predicts rain, M2 predicts rain, it rains respectively. We are given that X, Y are independent and $P(Z|X) = P(Z|Y) = 0.6$.

$$P(X \cap Z)/P(X) = P(Y \cap Z)/P(Y)$$

It could be that $P(X)=P(Y)=0.6$ and $Z=X \cap Y$. In that case $P(Z|A, B)=1$.

It could also be that $P(X)=P(Y)=0.4$ and $Z=(X \cap Y)'$. In that case $P(Z|A, B)=0$.

And anything in between is possible.

11. A disease test is advertised as being 91% accurate. If you have the disease you will test positive 99% of the time, and if you don't have the disease you will test negative 99% of the time. If 1% of all people have this disease and you feel positive, what is the probability that you actually have disease?

(a) 2/3

(b) 1/3

(c) 1/2

(d) 1/4

Solution:

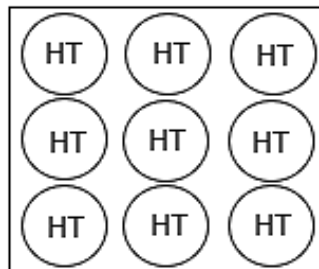
Let D be the event having disease + be the event that the testing positive.

Let $\bar{D}$ be the event not having disease.

We need to find

$$\begin{aligned}
 P(D/+) &= \frac{P(D \cap +)}{P(+)} \\
 &= \frac{P(D) \cdot P(+/D)}{P(D)P(+/D) + P(\bar{D})P(+/\bar{D})} \\
 &= \frac{1\% \cdot 99\%}{1\% \cdot 99\% + 99\% \cdot 1\%} \\
 &= \frac{99/100}{99/100 + 99/100} \\
 &= \frac{1}{2} \\
 &= 0.5
 \end{aligned}$$

12. Emma's coin box contains 8 fair, standard coins (head and tails) and 1 coin which has head on both sides. He selects a coin randomly and flips it 4 times, getting all heads. If he flips the coin again, what is the probability it will be heads?



(a) 1/6

(b) 1/3

(c) 5/6

(d) 2/3

Solution:

Let F and H_4 be the events of having the fair coin and flipping 4 heads respectively.

Let $\bar{F}$ be the event having unfair coin.

$$P(H_5) = P(F/H_4)P(H_4/F) + P(\bar{F}/H_4)P(H_4/\bar{F})$$

We need to find, $P(F/H_4)$ and $P(\bar{F}/H_4)$

$$P(\bar{F}/H_4) = 1 - P(F/H_4)$$

$$\begin{aligned} P(F/H_4) &= \frac{P(F \cap H_4)}{P(H_4)} \\ &= \frac{P(H_4/F) \cdot P(F)}{P(H_4/F) \cdot P(F) + P(H_4/\bar{F}) \cdot P(\bar{F})} \\ &= \frac{\frac{1}{16} \times \frac{8}{9}}{\frac{1}{16} \times \frac{8}{9} + 1 \times \frac{1}{9}} \\ &= \frac{1}{3} \end{aligned}$$

$$\therefore P(F/H_4) = \frac{2}{3}$$

$$P(H_5) = \frac{1}{3} \times \frac{1}{2} + \frac{2}{3} \times 1 = \frac{5}{6}$$

13. In general, the probability that it rains on Saturday is 25%. If it rains on Saturday the probability that it rains on Sunday is 50%. If it doesn't rain on Saturday, the probability that it rains on Sunday is 25%.

Given that it rained on Sunday, what is the probability that it rained on Saturday?

- | | |
|---------|-----------|
| (a) 40% | (b) 35% |
| (c) 60% | (d) 35.7% |

Solution:

It is given that

$$P(Su/Sa) = 50\%$$

$$\text{and } P(Su/\bar{Sa}) = 25\%$$

It is given that

$$P(Sa) = 25\%$$

$$\therefore P(\overline{Sa}) = 75\%$$

We need to find

$$\begin{aligned} P(Su/Sa) &= \frac{P(Sa \cap Su)}{P(Su)} \\ &= \frac{P(Sa) P(Su/Sa)}{P(Sa)P(Su/Sa) + P(\overline{Sa})P(Sa/\overline{Sa})} \\ &= \frac{25\% \times 50\%}{25\% \times 50\% + 75\% \times 25\%} \\ &= 40\% \end{aligned}$$

14. 1% of people have a rare cancer and there is a test of this cancer which is “90% accurate” that is

If you have the cancer there is a 90% chance the test will be positive.

If you don’t have the cancer there is a 90% chance the test will be negative.

If you take the test and positive, what is the approximate probability that you have the cancer?

(a) 10%

(b) 90%

(c) 50%

(d) None of these

Solution:

We need to find

$$\begin{aligned} P(C/+) &= \frac{P(C \cap +)}{P(+)} \\ &= \frac{P(C)P(+/C)}{P(C)P(+/C) + P(\overline{C})P(+/\overline{C})} \end{aligned}$$

Given data

$$P(C) = \frac{1}{100}$$

$$P(+/C) = \frac{90}{100}$$

$$\therefore P(C/+) = \frac{\frac{1}{100} \times \frac{9}{10}}{\frac{1}{100} \times \frac{9}{10} + \frac{99}{100} \times \frac{1}{10}}$$

$$\begin{aligned}
 &= \frac{9}{108} \\
 &= \frac{1}{12} \\
 &\approx 10\%
 \end{aligned}$$

15. A family has two children. Given that one of the children is a boy. What is the probability that both children are boys?

(a) $1/3$

(b) $1/2$

(c) $13/27$

(d) $11/27$

Solution:

There are 4 ways {bb, bg, gb, gg}

It is given that the family already has a boy.

$\therefore$ The probability of both children is $1/3$.

16. A family has two children. Given that one of the children is a boy and that he was born on a Tuesday. What is the probability that both children are boys?

Assume that the probability of being born on a particular day of the week is $1/7$ and is independent of whether the child is boy or girl?

(a) $1/3$

(b) $1/2$

(c) $13/27$

(d) $11/27$

Solution:

Let B be the event that the family has one child who is a boy on Tuesday.

A be the event that both children are boys.

$\therefore$ Given that there are 7 days of the week, there are 49 possible combinations for the days of the week the two boys were born on. 13 of these have boy who was born on a Tuesday.

$$P\left(\frac{B}{A}\right) = \frac{13}{49}$$

$$P(A) = \frac{1}{4}$$

We need to find $P\left(\frac{A}{B}\right)$

$$\begin{aligned} P\left(\frac{A}{B}\right) &= \frac{P(A \cap B)}{P(B)} \\ &= \frac{P(A)P\left(\frac{B}{A}\right)}{P(B)} \end{aligned}$$

$$P(B) = P(\text{one child who is a boy born on Tuesday})$$

$\therefore 14^2$ possible ways to select the gender and the day of the week the child was born on.

$\Rightarrow$ First child can be boy or girl and can be born on every any day of week.

$$7 + 7 = 2 \times 7 = 14$$

$\therefore$ Only one way this child can be born on tuesday.

$\Rightarrow$ This appears to the 2^{nd} child also again

$\therefore$ Only one way this child can be born on tuesday.

14 way

$$\therefore \text{Total number of way} = 14 \times 14 = 196$$

13^2 ways which don't have a boy born on Tuesday.

$\Rightarrow$ First child can be boy or girl who didn't born on Tuesday

$$14 - 1 = 13$$

$\Rightarrow$ Same way as first child

$$\therefore \text{Number of ways} = 14 - 1 = 13$$

$$\therefore \text{Total number of ways which don't have a boy born on Tuesday} = 13 \times 13 = 169$$

$$\therefore \text{Number of ways one child is a boy born on Tuesday} = 196 - 169 = 27$$

$$\therefore P(B) = \frac{27}{196}$$

$$\begin{aligned} P\left(\frac{A}{B}\right) &= \frac{P(A) \cdot P\left(\frac{B}{A}\right)}{P(B)} \\ &= \frac{\frac{13}{49} \times \frac{1}{4}}{\frac{27}{196}} \\ &= \frac{13}{27} \end{aligned}$$

17. Box 1 contains 1 red ball and 2 white balls. Box 2 contains 2 red balls and 1 white balls. One ball is drawn randomly from box 1 and transferred to box 2.

Then a ball is drawn randomly from box 2 and it is red. What is the probability that the transferred ball was white?

(a) $7/12$

(b) $3/4$

(c) $4/7$

(d) $1/2$

Solution:

Let W be that the transferred ball is white.

Let the event R be that the ball is drawn is red.

$$\therefore P(W) = \frac{2}{3} \text{ and } P\left(\frac{R}{W}\right) = \frac{2}{4} = \frac{1}{2}$$

It is given that the ball drawn is red.

∴ and $P(R)$ is dependent on which ball is transferred.

$$\begin{aligned}P(R) &= P(W)P\left(\frac{R}{W}\right) + P(\neg W)P\left(\frac{R}{\neg W}\right) \\&= \frac{2}{3} \times \frac{2}{4} + \frac{1}{3} \times \frac{3}{4} \\&= \frac{7}{12}\end{aligned}$$

we need to find,

$$\begin{aligned}P\left(\frac{W}{R}\right) &= \frac{P\left(\frac{R}{W}\right)P(W)}{P(R)} \\&= \frac{\frac{1}{2} \times \frac{2}{3}}{\frac{7}{12}} \\&= \frac{4}{7}\end{aligned}$$

18. There are four boxes. Each contains 2 balls the first box has a red and a white ball in it. The remaining three boxes each have two white balls in them.

A ball is picked at random from box 1 and put in box 2. Then a ball is picked at random from box 2 and put into 3. Then a ball is picked from box 3 and put into box 4. Finally a ball is picked from box 4.

The probability that the ball picked from box 2 is red, given that the final ball picked from box 4 is white is _____ [enter 3 decimal digits]

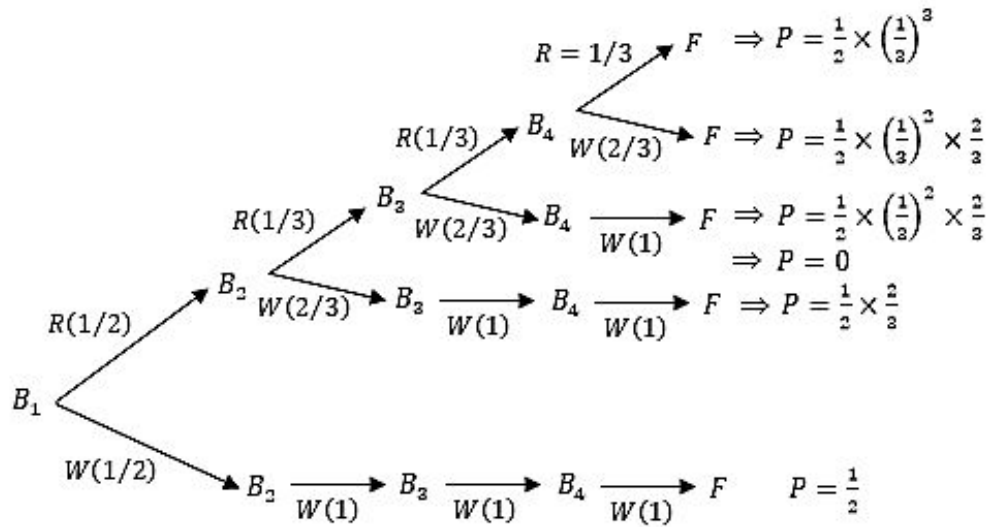
Solution:

Let R_2 = Probability that a red ball was picked from the second box

Let W_4 = Probability that a white ball was picked out from the fourth box

We need to find,

$$\begin{aligned}P\left(\frac{R_2}{W_4}\right) &= \frac{P(R_2 \cap W_4)}{P(W_4)} \\&= \frac{P\left(\frac{W_4}{R_2}\right) \times P(R_2)}{P(W_4)}\end{aligned}$$



$$\therefore P(W_4) = 1 - P(R_4)$$

$$= 1 - \frac{1}{2} \times \left(\frac{1}{3}\right)^3$$

$$= \frac{53}{54}$$

$$P(W_4/R_2) = \frac{2}{3} \times 1 + \frac{1}{3} \times \frac{2}{3}$$

$$= \frac{8}{9}$$

$$P(R_2) = \frac{1}{6}$$

$$P\left(\frac{R_2}{W_4}\right) = \frac{\frac{8}{9} \times \frac{1}{6}}{\frac{53}{54}}$$

$$= \frac{8}{53}$$

19. Harmoine granger is taking a question on a multiple choice test, she either knows the answer or guesses.

Let p be the probability that she knows the answer and $1 - p$ be the probability that she guesses.

Assume if she answers a question by guess, it will be correct with probability $1/m$, where m is the number of multiple choice alternatives.

What is the conditional probability that she knew the answer to a question given that she answered it correctly?

(a) $\frac{mp}{1+mp}$

$$(b) \frac{(m-1)p}{1+(m-1)p}$$

$$(c) \frac{mp}{1+(m-1)p}$$

$$(d) \frac{(m-1)p}{1+mp}$$

Solution:

Let c and k denote, respectively the events that she answers question correctly and the event that she actually knows the answer.

$$\begin{aligned} p\left(\frac{k}{c}\right) &= \frac{p(k \cap c)}{p(c)} \\ &= \frac{p\left(\frac{c}{k}\right)p(k)}{p\left(\frac{c}{k}\right)p(k) + p\left(\frac{c}{k'}\right)p(k')} \\ &= \frac{p}{p + \frac{1}{m}(1-p)} \end{aligned}$$

$p(k \cap c)$ is actually she knows the answer which is ' p '

$$= \frac{mp}{1+(m-1)p}$$

20. Consider a medical practitioner pondering the following dilemma: "If I am at least 80% certain that my patient has this disease, then I always recommend surgery, whereas if I am not quite certain, then I recommend additional tests that are expensive and sometimes painful. Now, initially I was only 60% certain that Emma had the disease. So I ordered the series A test, which always gives a positive when the patient has the disease and almost never does when she is healthy.

The test result was positive and I was all set to recommend surgery when Emma informed me, for the first time that she is a diabetic.

This information complicates matters because although it doesn't change my original 60% estimate of her chances of having disease, it does effect the interpretation of the results of the A test. This is so because the A test while never yielding a positive result when the patient is healthy, does unfortunately yield appositve result 30% of the time in the case of diabetic patients not suffering from the disease. Now what to do? More tests or immediate surgery?

(a) More tests

(b) Surgery

(c) Any one of this

(d) None of the above

Solution:

Let D denote the event that Emma has the disease.

E the event of a positive A test result, we need to find $P(\frac{D}{E})$

$$\begin{aligned} P(\frac{D}{E}) &= \frac{P(D \cap E)}{P(E)} \\ &= \frac{P(\frac{E}{D})P(D)}{P(\frac{E}{D})P(D) + P(\frac{E}{D'})P(D')} \\ &= \frac{1 \times 0.6}{1 \times 0.6 + 0.3 \times 0.4} \\ &= 0.833 \end{aligned}$$

∴ It is given 30% of the time in the case of diabetic patients not suffering from disease.

$$P(\frac{E}{D'}) = 30\%$$

$$\begin{aligned} P(D') &= 1 - P(D) \\ &= 1 - 60\% \\ &= 0.4 \end{aligned}$$

21. A letter is known to have come either from LONDON or CLIFTON; on the postmark on the two consecutive letters ON are legible. The probability that it come from LONDON is

- (a) $\frac{5}{17}$
- (b) $\frac{12}{17}$
- (c) $\frac{17}{30}$
- (d) $\frac{3}{5}$

Solution:

We need to find

$$\begin{aligned} P(\frac{LONDON}{ON}) &= \frac{P(LONDON \cap ON)}{P(ON)} \\ &= \frac{P(\frac{ON}{LONDON})P(LONDON)}{P(\frac{ON}{LONDON})P(LONDON) + P(\frac{ON}{CLIFTON})P(CLIFTON)} \end{aligned}$$

$$P(\frac{ON}{LONDON}) = \frac{2}{5} [5 \Rightarrow \text{Two consecutive letters can appear in 5 ways}]$$

$\{LO, ON, ND, DO, ON\}$ there is only one ways can ON legible]

$$P\left(\frac{ON}{CLIFTON}\right) = \frac{1}{6} [Two \text{ consecutive letters can appear in 6 ways}]$$

$\{CL, LI, IF, FT, TO, ON\}$ there is only one ways can ON legible]

$$\begin{aligned} & \cdot \cdot \frac{\frac{2 \times \frac{1}{2}}{\frac{5}{2}}}{\frac{2 \times \frac{1}{2} + \frac{1}{6} \times \frac{1}{2}} \\ &= \frac{\frac{1}{5}}{\frac{1}{5} + \frac{1}{12}} \\ &= \frac{\frac{1}{5}}{\frac{17}{12 \times 5}} \\ &= \frac{12}{17} \end{aligned}$$

22. A certain disease effects about 1 out of 10,000 people. There is a test to check whether the person has the disease. The test is quite accurate. In particular, we know that

- the probability that the result is positive (suggesting the person has the disease), given that the person does not have the disease, is only 2%.
- the probability that the test result is negative (suggesting the person does not have the disease), given that the person has the disease is only 1%.

A random person gets tested for the disease and the result comes back positive. What is the probability that the person has the disease?

- (a) < 50%
 (b) > 50%
 (c) 50%
 (d) 0% or 100%

Solution:

Let D be the event that the person has the disease, T be the event that the result be positive

$$P(D) = \frac{1}{10,000}$$

$$P\left(\frac{T}{D'}\right) = 0.02$$

$$P(T'/D) = 0.01$$

We want to compute $P(D/T)$

$$P\left(\frac{D}{T}\right) = \frac{P\left(\frac{T}{D}\right)P(D)}{P\left(\frac{T}{D}\right)P(D) + P\left(\frac{T}{D'}\right)P(D')}$$

$$= \frac{(1-0.01) \times 0.0001}{(1-0.01) \times 0.0001 + 0.02(1-0.0001)}$$

$$= 0.0049$$

$\therefore < 50\%$

23. In a market, they sell eggs in egg holders they share 10 of them in each. There is 60% chance, that all of the eggs are ok, 30% chance, that exactly 10 of them is broken, and 10% chance that exactly 2 of them are broken.

We buy an egg holder and after we grab our first egg, we are sad because it is broken, what is the probability that there is one more broken egg in our holder?

(a) $\frac{3}{5}$

(b) $\frac{1}{5}$

(c) $\frac{2}{5}$

(d) $\frac{1}{4}$

Solution:

Let B be the event where the first egg we observe is broken.

H_1 be the event our holder has one broken egg.

H_2 be the event where our holder has two broken eggs.

Since there's one broken egg out of ten in the H_1 case.

$$P\left(\frac{B}{H_1}\right) = \frac{1}{10}$$

$$\text{likewise } P\left(\frac{B}{H_2}\right) = \frac{2}{10}$$

$$\therefore P(B) = \frac{1}{10} \times \frac{5}{10} + \frac{2}{10} \times \frac{1}{10} = \frac{5}{100}$$

$$P\left(\frac{B}{H_1}\right)P(H_1) + P\left(\frac{B}{H_2}\right)P(H_2)$$

We need to find,

$$P\left(\frac{H_2}{B}\right) = \frac{P\left(\frac{B}{H_2}\right)P(H_2)}{P(B)}$$

$$= \frac{\frac{2}{10} \times \frac{1}{10}}{\frac{5}{100}}$$

$$= \frac{2}{5}$$

24. I'm on a farm with six cows; three are white, two are white, two are black and one is completely black on one side and completely white on the other. I see one cow from one side which appears to be black. What's the probability that the cow's black?

Solution:

Let B be the event that the cow I observe is completely black, and S be the event that atleast one side is black. S is given, so I need to find $P(B/S)$.

$$P\left(\frac{B}{S}\right) = \frac{P\left(\frac{S}{B}\right) * P(B)}{P(S)}$$

$P\left(\frac{S}{B}\right) = 1$, since if the entire cow is black then atleast one side is black. $P(B) = 1/3$ since there are two black cows among a group of six.

$P(S) = 5/12$ since there are 12 sides you can see, all equally likely, and 5 of the sides are black. So, $P(S) = 5/12$.

$$\text{This yields } P\left(\frac{B}{S}\right) = \frac{\left(\frac{1}{3}\right)}{\left(\frac{5}{12}\right)} = \frac{4}{5}$$

25. A person has two boxes A and B. First one A has 4 white balls and 5 black balls and the second or B has 5 white balls and 4 black balls. This person takes randomly one ball from the first box and put into the second box. After that he takes a ball from the second box. Find the probability of taking balls of the same color in this process?

Solution:

The probability of choosing white from the first box and then from the second box is:

$$\left(\frac{4}{9}\right) \cdot \left(\frac{6}{10}\right) = \frac{24}{90}$$

The probability of choosing black from the first box and then from the second box is:

$$\left(\frac{5}{9}\right) \cdot \left(\frac{5}{10}\right) = \frac{25}{90}$$

So, the probability of choosing the same color is:

$$\frac{24}{90} + \frac{25}{90} = \frac{49}{90}$$

26. Prof X is crossing the atlantic ocean on a plane, on his way to a conference. The captain has just announced that on unusual engine taught has been signaled by the plane is computer. This indicates a fault that only occurs once in 10,000 flights. If the fault report is true then there is a 70% chance the plane will have to crash land in the ocean, which means certain death for the

passengers. However the reports are completely reliable. There is a 2% chance of a false positive; and there is a 1% chance of the same fault occurring without the computer flagging the error report.

Solution:

Baye's law :

$$P\left(\frac{A}{B}\right) = \frac{P\left(\frac{B}{A}\right) \cdot P(A)}{P(B)}$$

$$P\left(\frac{A}{B}\right) = \frac{P\left(\frac{B}{A}\right) \cdot P(A)}{P(B)}$$

$$P\left(\frac{Fault}{Positive\ Indication}\right) = \frac{P\left(\frac{Positive\ Indication}{Fault}\right) \cdot P(Fault)}{P(Positive\ Indication)}$$

$$P\left(\frac{Positive\ Indication}{Fault}\right) = 0.99$$

$$P(Fault) = 0.0001$$

P(Positive Indication) is a little trickier. So, lets make this table.

	Fault	NoFault	Total
Positive	$0.99 * 0.0001$	$0.02 * 0.9999$	0.020097
Negative	$0.01 * 0.0001$	$0.98 * 0.9999$	0.979902
Total	0.00001	0.9999	

$$P\left(\frac{Fault}{Positive\ Indication}\right) = \frac{0.000099}{0.020097} = 0.0049$$

$$P(Survival) = 1 - P(Death) = (1 - 0.0049) * 0.70 = 0.9965$$

